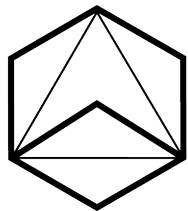




Golf Cart bring-up progress



THE
AUTWARE
FOUNDATION

National Taiwan University

Autware LSV meeting, August 2026

國立臺灣大學



The vehicle

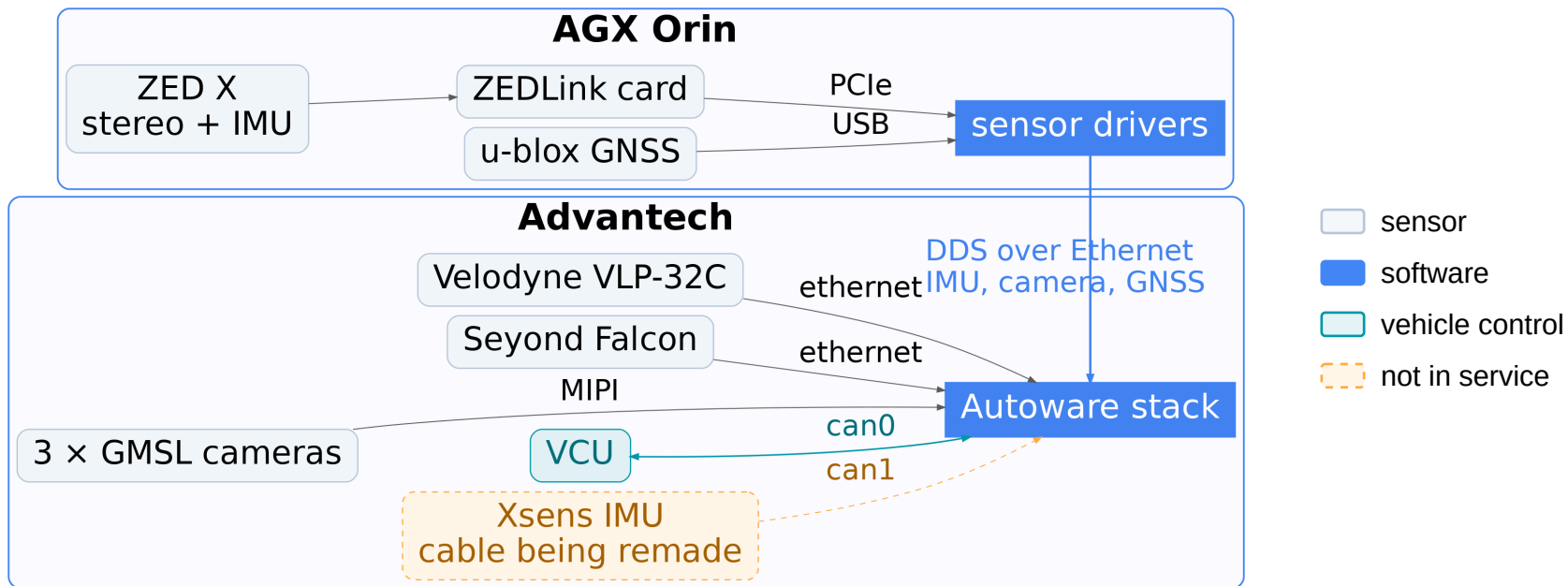


A low-speed campus vehicle on a **Turing Drive VCU**, with an Autoware stack across **two machines**.

Bring-up in six steps:

1. Sensors ready
2. Two-host system ready
3. Vehicle interface in progress
4. Map preparation ready
5. Data collection done, data flawed
6. Autonomous run not started

Where we are



Sensors are up

Every sensor on the vehicle publishes, on both machines.

Sensor	Host	Feeds
Velodyne VLP-32C	Advantech	NDT scan matching, perception
Seyond Falcon	Advantech	perception
3 GMSL cameras	Advantech	perception, recording
ZED X stereo + IMU	Orin	the IMU the stack runs on today
u-blox GNSS	Orin	pose initialisation
Turing Drive VCU	Advantech	velocity, gear, subsystem states

Two machines, one stack

ROS 2 dropped ROS 1's machine tag, so orchestration is ours. It works.

One instance, nothing left behind

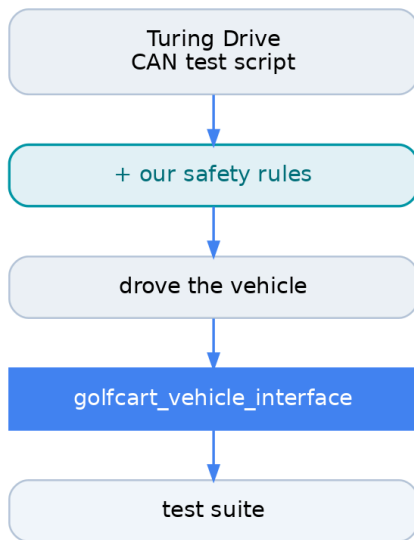
- **play_launch** cleans up orphans when the launch dies, and starts the stack in **20 s** against **60 s** for `ros2 launch`
- **systemd user units** around it: singleton by construction, `KillMode=control-group` takes the whole tree down

One command, either machine

- one CycloneDDS profile per role, plus a config/host marker file
- `scripts/env.sh` reads it, and every shell and every unit sources it
- so a terminal on either box is correct the moment it opens. Nobody has to remember which machine they are on

`just launch-all` brings both hosts up and returns. `just stop-all` takes them down.

The interface drives the cart



Not written from a specification. Grown from the vendor's own test code, and every step exists in the repo.

The safety rules are ours:

- target speed can never go negative, reverse is gear R
- gear P pins speed and angle to zero, re-applied every cycle
- ESTOP release is never a key the terminal cannot send

A test suite sits on top: keyboard control, a command service, and trajectory replay.

Data collection works



Three runs at NTU. **Both hosts record separately**: each writes the topics for the devices it owns, and the bags merge afterwards.

Only **first-hand driver output** is recorded. Anything a node computed is left out, so replay recomputes it with today's parameters rather than the ones frozen at record time.

Replay is one command: just `ntu-test run`. Bag paused for `/clock`, then `stack`, `RViz`, initial pose, in the only order that works.

What bit us

Three things worth knowing before you copy this.

- **The vendor's oToCam setup gives you video and kills every USB port.** We worked around it, and now have both. The .ko files stay ABI-bound to kernel 5.15.148-tegra, so a JetPack OTA silently undoes it again.
- **The Xsens IMU cable broke, and is being remade.** Meanwhile the stack runs on the **ZED X built-in IMU**, which sits on the other machine and crosses the network at 100 Hz.
- **The u-blox GNSS is on the Orin**, because the Advantech is short of USB ports. So a second localization input is remote too, for an unrelated reason.

The machine is at its limit

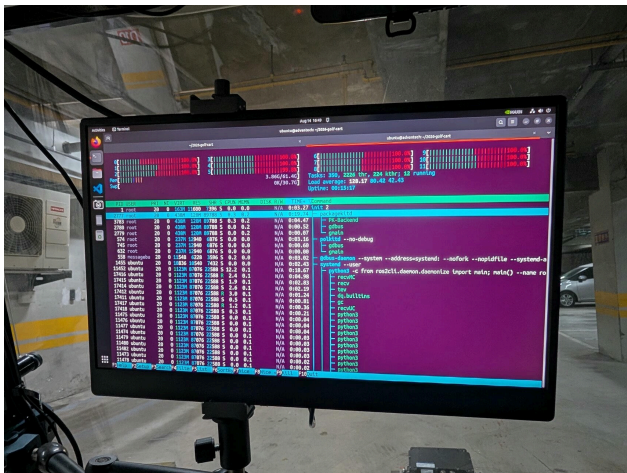
The GMSL cameras emit **UYVY** at 1920x1280, 30 fps, three of them. The consuming nodes want **RGB or JPEG**, so something must convert. We tried **nvvidconv**. It still costs us CPU, per camera.

That is one reason the box runs hot enough to need a fan held against it by hand, and one reason there are **two machines** at all.

Fix in progress: `gmslcam`, to take that stage out.



The bill for that speed



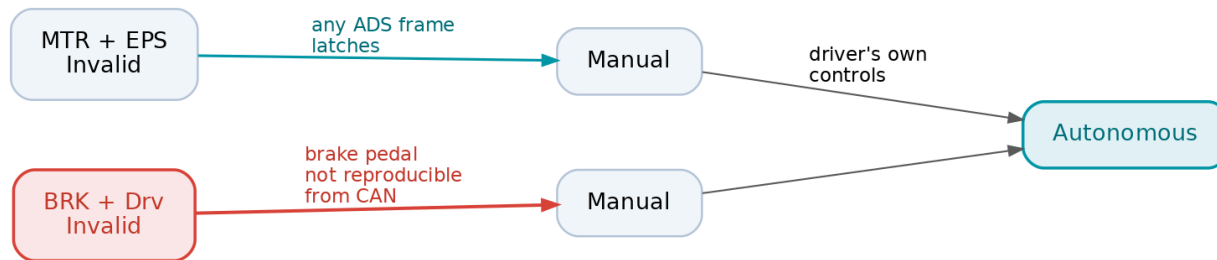
play_launch is fast *because* it spawns as fast as it can. On a box already at its limit, that is a thundering herd, and the machine locks up hard enough to need a power cycle.

Pacing the spawns is the obvious fix. We measured it losing: about 10% fewer runnable tasks for more than double the startup time, spending the exact advantage we adopted the tool for.

What ships is a **1 GiB MemAvailable floor**. Nothing on a healthy boot; holds back only when the machine is about to die.

Gate on the resource that runs out, not on the rate.

Engage is the VCU's decision, and one door we cannot open

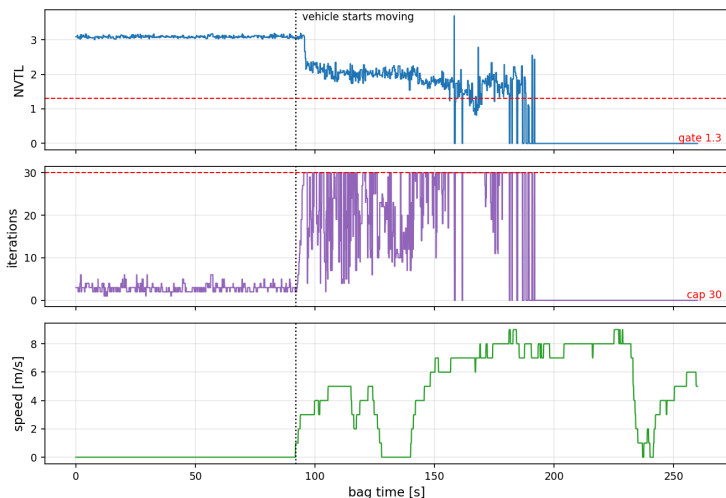


There is no `vehicle_cmd_gate` external selector. The interface commands nothing until all four subsystems report autonomous. **No service call can force it.**

After a VCU restart, **BRK and Drv come up Invalid**, and only a brake-pedal press was seen to clear them. Not reproducible from CAN.

Unattended autonomous start-up is blocked. A vendor question, not more software on our side.

NDT still breaks



We tuned it and measured it on **scan-to-map residual**, not on the NVTL score. The change that helped accuracy most actually **lowered** NVTL.

It converges parked, to 0.14 m. It **degrades a few seconds after the vehicle starts moving**.

The cause is upstream of NDT, in the recording: 45% of scans are fragments of a revolution, and scans arrive **376 ms stale**.

No raw LiDAR packets were recorded, so these bags cannot be re-decoded. The next run records them.

Status

Step	State	What stands in the way
Sensors	ready	fragmented LiDAR scans; the Xsens cable is being remade
Two-host system	ready	none
Vehicle interface	in progress	the VCU will not enter its autonomous state from CAN alone
Map preparation	ready	PCD and lanelet2 from Turing Drive, downsampled at runtime
Data collection	done, data flawed	the bags carry the sensor defects above
Autonomous run	not started	gated on the vehicle interface blocker

Next

- **Ask Turing Drive what clears the Invalid brake state after a restart.** Everything else on the autonomous run waits behind this.
- **Record raw LiDAR packets**, and check the VLP-32C rotation configuration against the hardware. Then re-run the localization work on data that is not already broken.
- **Migrate the sensor kit to gmslcam**, and take the per-camera CPU conversion out.

TSN is a separate track, happy to cover it if there is interest.